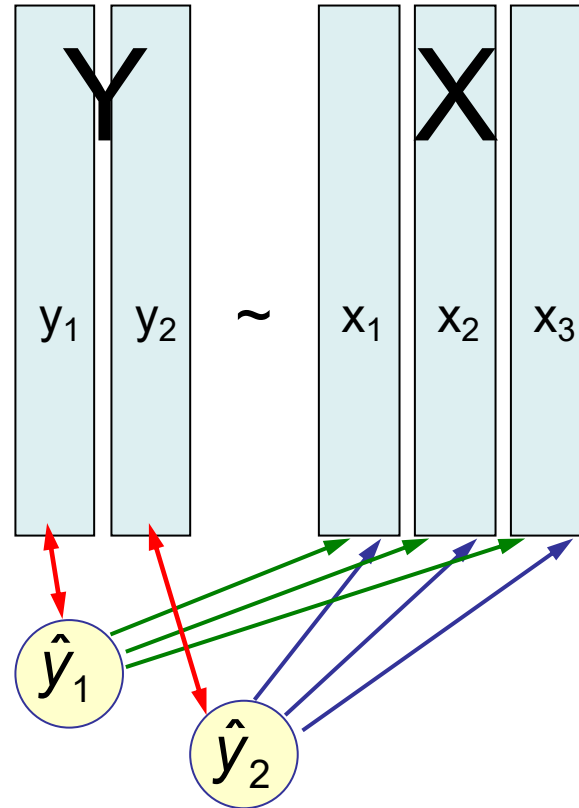
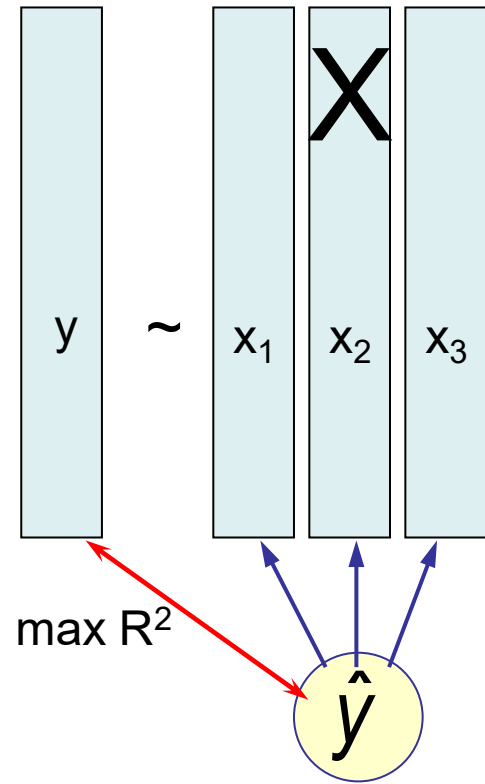
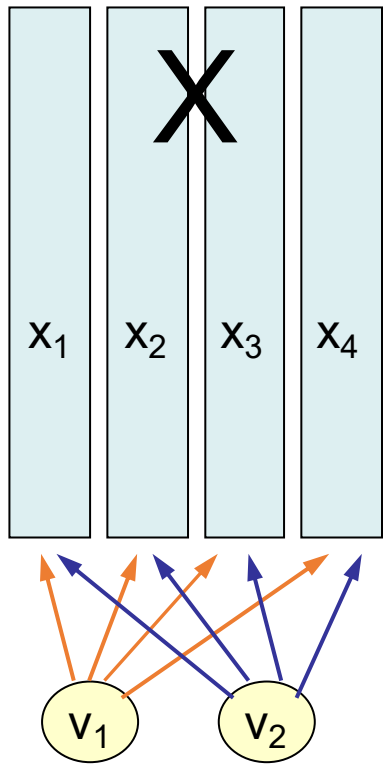
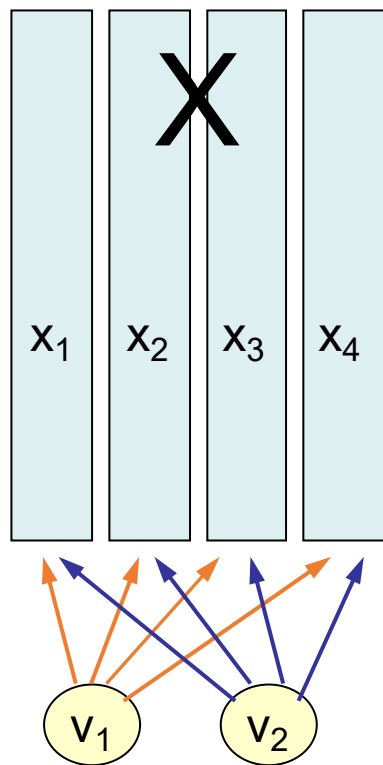
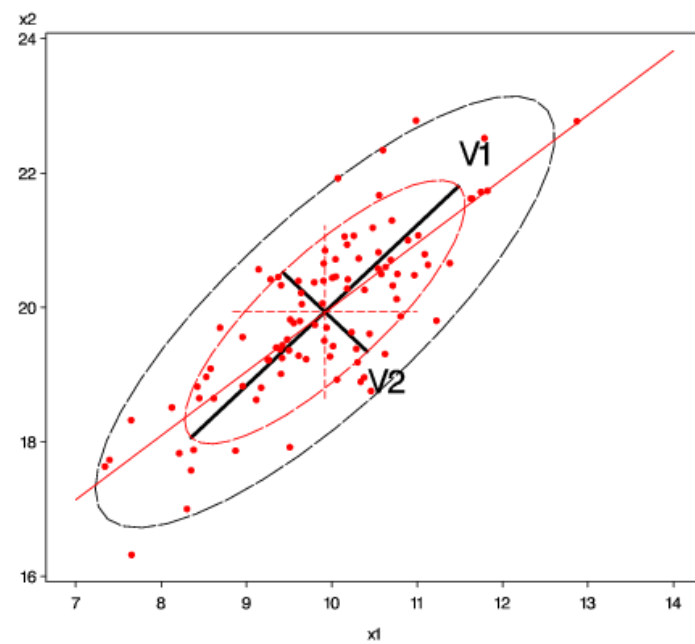


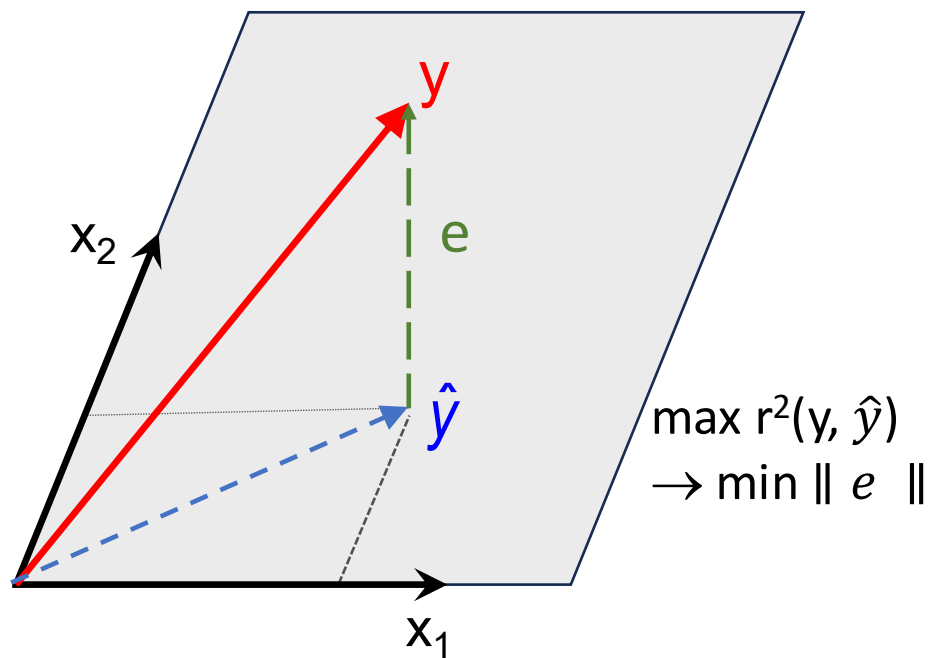
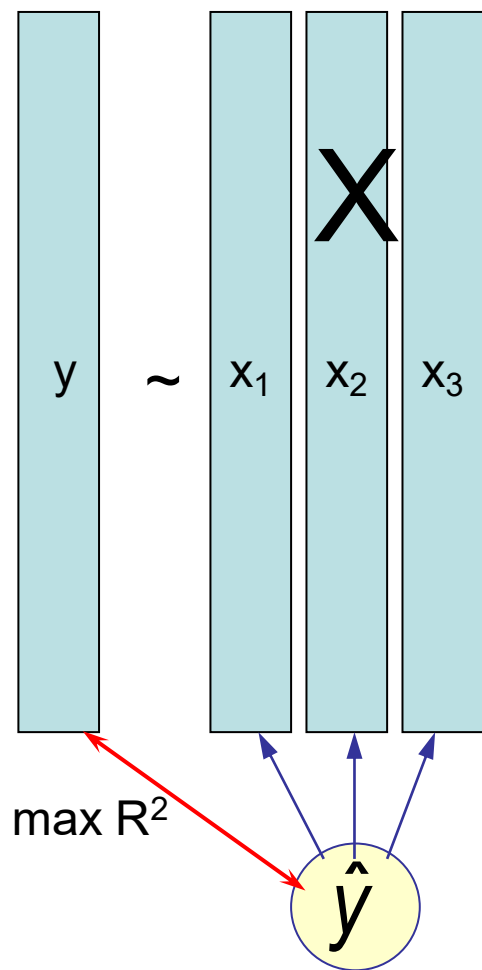


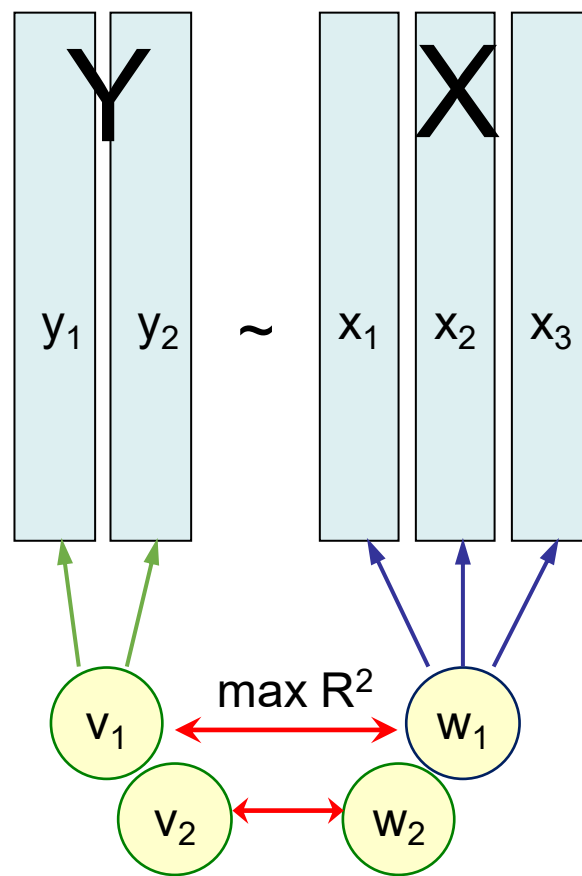
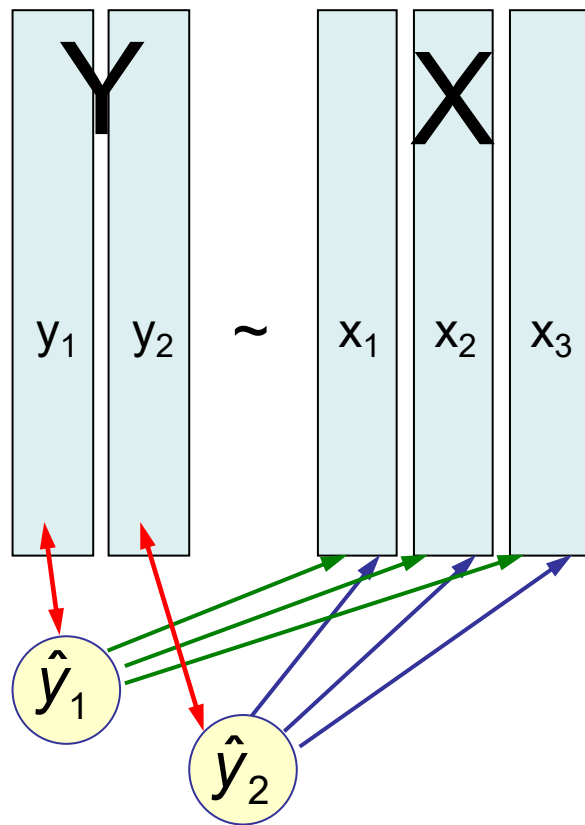
PCA



PCA: Linear combinations to maximize variance





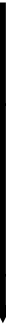


Response variables: $\mathbf{Y} = (y_1, \dots, y_q)$

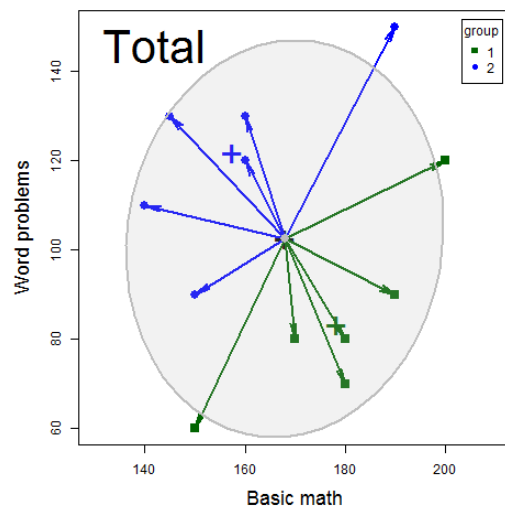
Predictor variables: $\mathbf{X} = (x_1, \dots, x_p)$

		Quantitative		Discrete	
		q=1	q>1	q=1	q>1
Quantitative	p=1	Simple regression	Multivariate regression	Simple logistic regression	
	p>1	Multiple regression	Multivariate regression Canonical corr. Partial corr.	Mult. logistic regression Discriminant analysis	Multivariate logistic regression
Discrete	p=1	t-test 1-way ANOVA	Hotelling T ² 1-way MANOVA	Simple χ^2	Loglinear models
	p>1	Factorial ANOVA	Factorial MANOVA	Logit models Loglinear models	Loglinear models

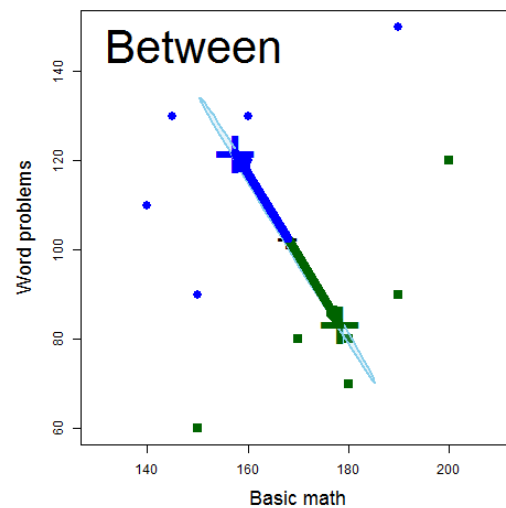
	Regression	ANOVA
Dependent (response)	Quantitative	Quantitative
Independent (predictors)	Quantitative	Discrete factors
Concepts, statistics	Terms: X_1, X_2 Interactions: $X_1 * X_2$ Linear hypotheses R^2 , coefficients	Main effects: A, B Interactions: A*B Contrasts F stats, factor effects

	Dependent variables	
Independent variables	1 Quantitative $y = X \beta$	2+ Quantitative $Y = X B$
Quantitative	Regression 	Multivariate regression 
Categorical	ANOVA 	MANOVA 
Both	Reg. w/ dummy vars ANCOVA Homogeneity of regression 	General MLM Homogeneity of regression MANCOVA 

$$(y_{ij} - \bar{y}_{..}) = (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$$



=



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